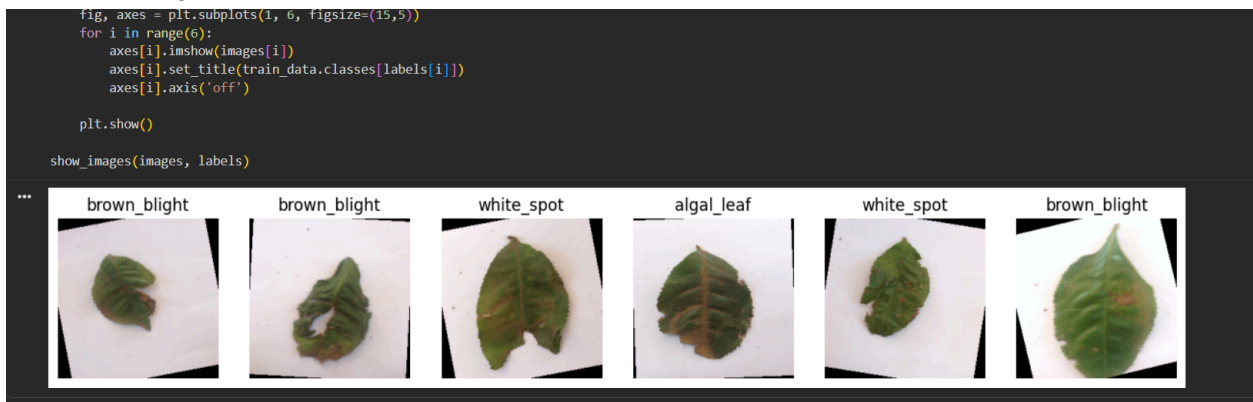


Split and Count

Split	Class	Count
train	brown_blight	79
train	algal_leaf	79
train	white_spot	99
	SUBTOTAL	257
val	brown_blight	17
val	algal_leaf	17
val	white_spot	21
	SUBTOTAL	55
test	brown_blight	17
test	algal_leaf	17
test	white_spot	22
	SUBTOTAL	56
TOTAL		368

Visualize images



Training Model 1 : VGG16

```
... Using device: cpu
Downloading: "https://download.pytorch.org/models/vgg16-397923af.pth" to /root/.cache/torch/hub/checkpoints/vgg16-397923af.pth
100%|██████████| 528M/528M [00:05<00:00, 94.3MB/s]
🔥 Training started...

Epoch 1/10
Train Loss: 1.0698 | Train Acc: 0.4280
Val Loss: 0.6773 | Val Acc: 0.8545
-----
Epoch 2/10
Train Loss: 0.8434 | Train Acc: 0.6576
Val Loss: 0.6953 | Val Acc: 0.6909
-----
Epoch 3/10
Train Loss: 0.7530 | Train Acc: 0.7821
Val Loss: 0.4114 | Val Acc: 0.8909
-----
🔴 Early stopping triggered
✅ Best model loaded (highest validation accuracy)

✅ Test Accuracy: 0.8750
```

Training Model 2 : ResNet50

```
Downloading: "https://download.pytorch.org/models/resnet50-11ad3fa6.pth" to /root/.cache/torch/hub/checkpoints/resnet50-11ad3fa6.pth
100%|██████████| 97.8M/97.8M [00:00<00:00, 164MB/s]
🔥 Training started...

Epoch 1/10
Train Loss: 1.0329 | Train Acc: 0.4591
Val Loss: 0.8373 | Val Acc: 0.6909
-----
Epoch 2/10
Train Loss: 0.6896 | Train Acc: 0.7860
Val Loss: 0.3015 | Val Acc: 0.9455
-----
Epoch 3/10
Train Loss: 0.3270 | Train Acc: 0.9105
Val Loss: 0.1974 | Val Acc: 0.9455
-----
🛑 Early stopping triggered
✅ Best model loaded (highest validation accuracy)

✅ Test Accuracy: 0.8929
```

Performance Comparison

```
import pandas as pd

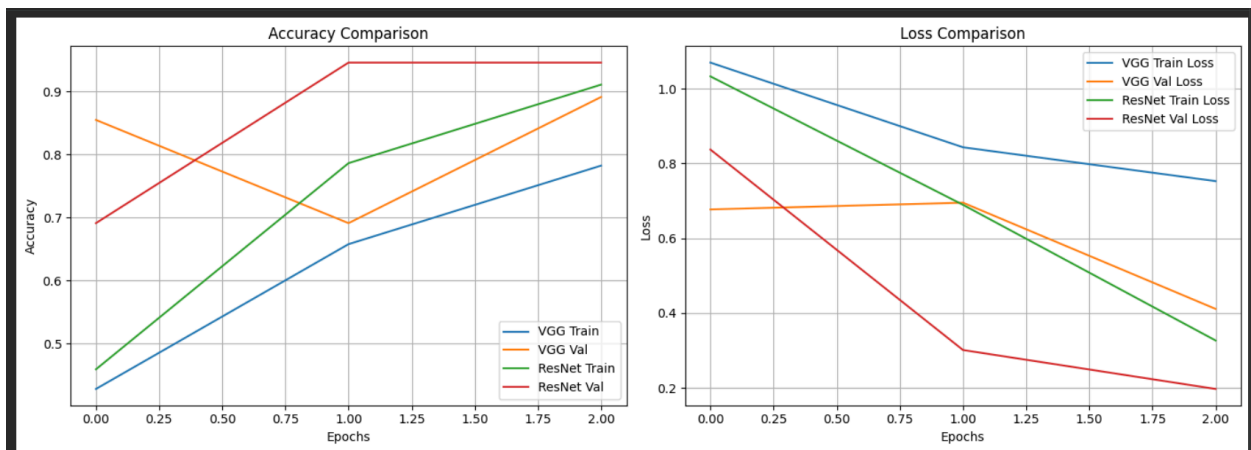
data = {
    "Model": ["VGG16", "ResNet50"],
    "Train Accuracy": [train_accs[-1], resnet_train_accs[-1]],
    "Validation Accuracy": [val_accs[-1], resnet_val_accs[-1]],
    "Test Accuracy": [vgg_test_acc, resnet_test_acc]
}

df = pd.DataFrame(data)

print(df)
```

	Model	Train Accuracy	Validation Accuracy	Test Accuracy
0	VGG16	0.782101	0.890909	0.875000
1	ResNet50	0.910506	0.945455	0.892857

Accuracy and Loss Comparison

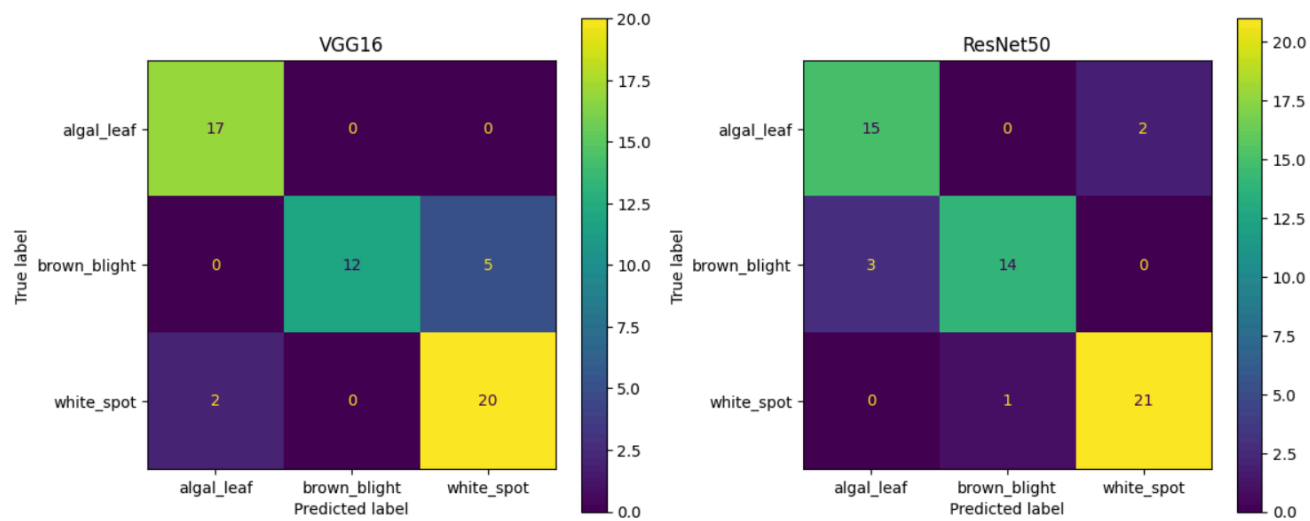


Classification Report

VGG16 Classification Report:				
	precision	recall	f1-score	support
algal_leaf	0.89	1.00	0.94	17
brown_blight	1.00	0.71	0.83	17
white_spot	0.80	0.91	0.85	22
accuracy			0.88	56
macro avg	0.90	0.87	0.87	56
weighted avg	0.89	0.88	0.87	56

ResNet50 Classification Report:				
	precision	recall	f1-score	support
algal_leaf	0.83	0.88	0.86	17
brown_blight	0.93	0.82	0.88	17
white_spot	0.91	0.95	0.93	22
accuracy			0.89	56
macro avg	0.89	0.89	0.89	56
weighted avg	0.90	0.89	0.89	56

Visualize Confusion Matrices



Predicting a sample image

...

True Label: brown_blight



True Class: brown_blight
Predicted Class: brown_blight
Confidence: 0.9443

The image shows a single green leaf with a distinct brown, necrotic area at the bottom, which is characteristic of brown blight. The leaf is set against a light background. The prediction results are displayed below the image, showing a high confidence score of 0.9443 for the predicted class 'brown_blight'.